Полный отчет анализа моделей

Сводные результаты

Серия	Модель	Средняя MAE	Средняя RMSE	Средняя sMAPE	Среднее время обучения	Лучшая модель
C	LSTM (128u, 12w, 100e, 16b)	0.5663	1.2103	0.5572%	16.84 сек	
C	LSTM (128u, 12w, 100e, 32b)	0.6139	1.2199	0.6044%	14.75 сек	
C	LSTM (128u, 12w, 100e, 64b)	0.5361	1.2080	0.5272%	14.27 сек	
C	LSTM (128u, 12w, 200e, 16b)	0.5609	1.2030	0.5518%	26.62 сек	
C	LSTM (128u, 12w, 200e, 32b)	0.6725	1.2501	0.6625%	22.54 сек	
C	LSTM (128u, 12w, 200e, 64b)	0.5929	1.2028	0.5836%	20.28 сек	
C	LSTM (128u, 12w, 50e, 16b)	0.5632	1.2171	0.5541%	12.07 сек	
C	LSTM (128u, 12w, 50e, 32b)	0.5242	1.2177	0.5153%	11.07 сек	
C	LSTM (128u, 12w, 50e, 64b)	0.5356	1.2151	0.5267%	10.88 сек	
C	LSTM (128u, 18w, 100e, 16b)	0.5658	1.1841	0.5566%	18.98 сек	
C	LSTM (128u, 18w, 100e, 32b)	0.5940	1.2127	0.5846%	16.00 сек	
C	LSTM (128u, 18w, 100e, 64b)	0.5507	1.2066	0.5417%	14.62 сек	
C	LSTM (128u, 18w, 200e, 16b)	0.7144	1.2587	0.7034%	30.56 сек	
C	LSTM (128u, 18w, 200e, 32b)	0.6395	1.2087	0.6297%	24.70 сек	
C	LSTM (128u, 18w, 200e, 64b)	0.5758	1.1966	0.5666%	22.22 сек	
C	LSTM (128u, 18w, 50e, 16b)	0.5563	1.2066	0.5472%	13.28 сек	
C	LSTM (128u, 18w, 50e, 32b)	0.5414	1.2069	0.5325%	27.19 сек	
C	LSTM (128u, 18w, 50e, 64b)	0.5536	1.2208	0.5446%	11.47 сек	
C	LSTM (128u, 24w, 100e, 16b)	0.6669	1.1832	0.6565%	20.26 сек	
C	LSTM (128u, 24w, 100e, 32b)	0.6477	1.1844	0.6375%	17.00 сек	
C	LSTM (128u, 24w, 100e, 64b)	0.5278	1.2093	0.5189%	15.68 сек	
C	LSTM (128u, 24w, 200e, 16b)	0.7438	1.2424	0.7329%	34.65 сек	
C	LSTM (128u, 24w, 200e, 32b)	0.6309	1.2522	0.6209%	27.29 сек	
C	LSTM (128u, 24w, 200e, 64b)	0.6637	1.2306	0.6534%	24.96 сек	
C	LSTM (128u, 24w, 50e, 16b)	0.5587	1.2214	0.5496%	14.44 сек	
C	LSTM (128u, 24w, 50e, 32b)	0.5678	1.2085	0.5587%	12.33 сек	
C	LSTM (128u, 24w, 50e, 64b)	0.5707	1.2228	0.5615%	11.18 сек	
C	LSTM (128u, 3w, 100e, 16b)	0.6390	1.2376	0.6293%	14.00 сек	
C	LSTM (128u, 3w, 100e, 32b)	0.5644	1.2222	0.5553%	12.99 сек	

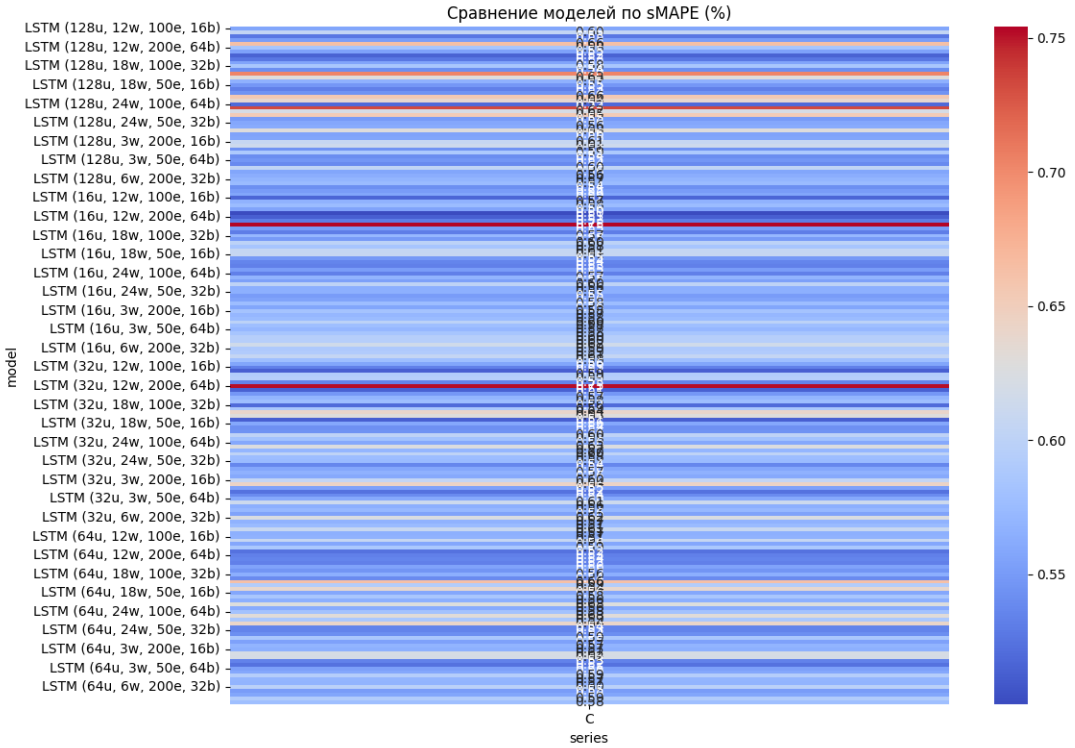
C	LSTM (128u, 3w, 100e, 64b)	0.5665	1.2227	0.5574%	12.73 сек	
C	LSTM (128u, 3w, 200e, 16b)	0.6158	1.2322	0.6063%	20.52 сек	
C	LSTM (128u, 3w, 200e, 32b)	0.6219	1.2335	0.6123%	19.86 сек	
C	LSTM (128u, 3w, 200e, 64b)	0.5530	1.2204	0.5440%	17.71 сек	
C	LSTM (128u, 3w, 50e, 16b)	0.5974	1.2286	0.5880%	24.44 сек	
C	LSTM (128u, 3w, 50e, 32b)	0.5499	1.2200	0.5409%	12.93 сек	
C	LSTM (128u, 3w, 50e, 64b)	0.5556	1.2201	0.5466%	10.97 сек	
C	LSTM (128u, 6w, 100e, 16b)	0.5472	1.2212	0.5382%	15.80 сек	
C	LSTM (128u, 6w, 100e, 32b)	0.6126	1.2326	0.6031%	13.81 сек	
C	LSTM (128u, 6w, 100e, 64b)	0.5663	1.2238	0.5572%	12.41 сек	
C	LSTM (128u, 6w, 200e, 16b)	0.5729	1.2244	0.5637%	22.27 сек	
C	LSTM (128u, 6w, 200e, 32b)	0.5787	1.2259	0.5695%	19.40 сек	
C	LSTM (128u, 6w, 200e, 64b)	0.5831	1.2267	0.5739%	17.88 сек	
C	LSTM (128u, 6w, 50e, 16b)	0.5502	1.2215	0.5412%	12.29 сек	
C	LSTM (128u, 6w, 50e, 32b)	0.5598	1.2227	0.5508%	12.41 сек	
C	LSTM (128u, 6w, 50e, 64b)	0.5670	1.2242	0.5579%	10.41 сек	
C	LSTM (16u, 12w, 100e, 16b)	0.5263	1.2030	0.5174%	17.14 сек	
C	LSTM (16u, 12w, 100e, 32b)	0.5763	1.2238	0.5671%	15.06 сек	
C	LSTM (16u, 12w, 100e, 64b)	0.5845	1.2239	0.5753%	13.43 сек	
C	LSTM (16u, 12w, 200e, 16b)	0.5644	1.1861	0.5552%	25.73 сек	
C	LSTM (16u, 12w, 200e, 32b)	0.5104	1.2098	0.5017%	23.63 сек	+
C	LSTM (16u, 12w, 200e, 64b)	0.5222	1.2106	0.5134%	19.97 сек	
C	LSTM (16u, 12w, 50e, 16b)	0.5388	1.2199	0.5299%	12.62 сек	
C	LSTM (16u, 12w, 50e, 32b)	0.7650	1.2765	0.7541%	12.11 сек	
C	LSTM (16u, 12w, 50e, 64b)	0.5596	1.2187	0.5505%	13.25 сек	
C	LSTM (16u, 18w, 100e, 16b)	0.5393	1.2177	0.5304%	18.30 сек	
C	LSTM (16u, 18w, 100e, 32b)	0.5808	1.2206	0.5715%	14.93 сек	
C	LSTM (16u, 18w, 100e, 64b)	0.5582	1.2213	0.5492%	13.73 сек	
C	LSTM (16u, 18w, 200e, 16b)	0.6092	1.2184	0.5997%	27.91 сек	
C	LSTM (16u, 18w, 200e, 32b)	0.5937	1.2213	0.5844%	22.21 сек	
C	LSTM (16u, 18w, 200e, 64b)	0.6161	1.2290	0.6066%	20.11 сек	
C	LSTM (16u, 18w, 50e, 16b)	0.6170	1.2327	0.6075%	12.84 сек	
C	LSTM (16u, 18w, 50e, 32b)	0.5580	1.2210	0.5490%	13.43 сек	
C	LSTM (16u, 18w, 50e, 64b)	0.5448	1.2183	0.5359%	11.40 сек	
C	LSTM (16u, 24w, 100e, 16b)	0.5414	1.2188	0.5324%	20.86 сек	

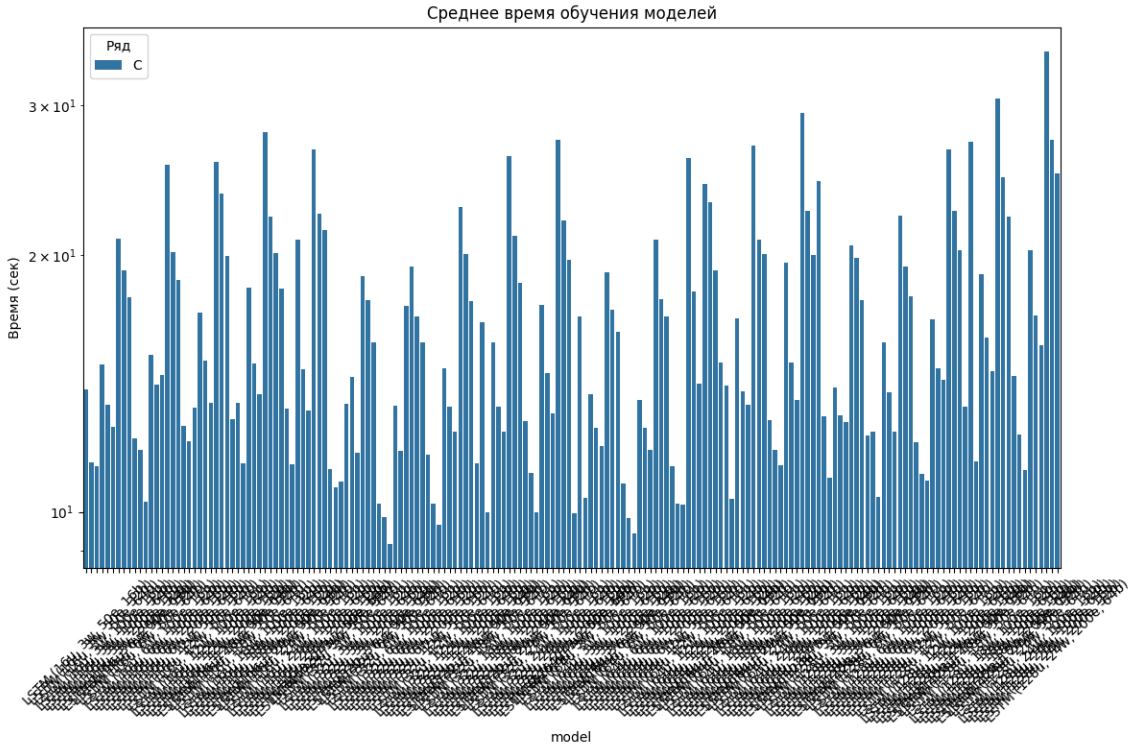
C	LSTM (16u, 24w, 100e, 32b)	0.5602	1.2228	0.5512%	14.69 сек	
C	LSTM (16u, 24w, 100e, 64b)	0.5419	1.2200	0.5329%	13.15 сек	
C	LSTM (16u, 24w, 200e, 16b)	0.5773	1.2012	0.5680%	26.59 сек	
C	LSTM (16u, 24w, 200e, 32b)	0.5626	1.2186	0.5535%	22.35 сек	
C	LSTM (16u, 24w, 200e, 64b)	0.6101	1.2326	0.6007%	21.41 сек	
C	LSTM (16u, 24w, 50e, 16b)	0.5736	1.2244	0.5645%	18.27 сек	
C	LSTM (16u, 24w, 50e, 32b)	0.5744	1.2234	0.5652%	13.23 сек	
C	LSTM (16u, 24w, 50e, 64b)	0.5592	1.2204	0.5502%	11.36 сек	
C	LSTM (16u, 3w, 100e, 16b)	0.5644	1.2219	0.5553%	14.88 сек	
C	LSTM (16u, 3w, 100e, 32b)	0.5854	1.2250	0.5762%	13.35 сек	
C	LSTM (16u, 3w, 100e, 64b)	0.5668	1.2220	0.5577%	12.58 сек	
C	LSTM (16u, 3w, 200e, 16b)	0.5909	1.2268	0.5816%	20.88 сек	
C	LSTM (16u, 3w, 200e, 32b)	0.5800	1.2247	0.5707%	19.21 сек	
C	LSTM (16u, 3w, 200e, 64b)	0.5844	1.2248	0.5752%	17.84 сек	
C	LSTM (16u, 3w, 50e, 16b)	0.6091	1.2300	0.5997%	13.93 сек	
C	LSTM (16u, 3w, 50e, 32b)	0.5876	1.2261	0.5784%	11.43 сек	
C	LSTM (16u, 3w, 50e, 64b)	0.5804	1.2250	0.5712%	11.32 сек	
C	LSTM (16u, 6w, 100e, 16b)	0.5932	1.2291	0.5838%	15.28 сек	
C	LSTM (16u, 6w, 100e, 32b)	0.6047	1.2311	0.5953%	14.11 сек	
C	LSTM (16u, 6w, 100e, 64b)	0.6045	1.2311	0.5951%	14.48 сек	
C	LSTM (16u, 6w, 200e, 16b)	0.6249	1.2357	0.6153%	25.56 сек	
C	LSTM (16u, 6w, 200e, 32b)	0.5987	1.2296	0.5893%	20.18 сек	
C	LSTM (16u, 6w, 200e, 64b)	0.6006	1.2306	0.5912%	18.69 сек	
C	LSTM (16u, 6w, 50e, 16b)	0.6148	1.2333	0.6053%	12.19 сек	
C	LSTM (16u, 6w, 50e, 32b)	0.5873	1.2279	0.5781%	11.83 сек	
C	LSTM (16u, 6w, 50e, 64b)	0.5666	1.2225	0.5575%	10.28 сек	
C	LSTM (32u, 12w, 100e, 16b)	0.5420	1.1966	0.5330%	14.75 сек	
C	LSTM (32u, 12w, 100e, 32b)	0.5220	1.2032	0.5131%	13.29 сек	
C	LSTM (32u, 12w, 100e, 64b)	0.6018	1.2278	0.5924%	12.43 сек	
C	LSTM (32u, 12w, 200e, 16b)	0.6038	1.2157	0.5944%	22.77 сек	
C	LSTM (32u, 12w, 200e, 32b)	0.5438	1.2040	0.5348%	20.06 сек	
C	LSTM (32u, 12w, 200e, 64b)	0.7626	1.2560	0.7517%	17.66 сек	
C	LSTM (32u, 12w, 50e, 16b)	0.5228	1.2112	0.5140%	11.67 сек	
C	LSTM (32u, 12w, 50e, 32b)	0.5565	1.2202	0.5474%	10.24 сек	
C	LSTM (32u, 12w, 50e, 64b)	0.5788	1.2222	0.5696%	9.65 сек	

C	LSTM (32u, 18w, 100e, 16b)	0.5869	1.2112	0.5776%	15.83 сек	
C	LSTM (32u, 18w, 100e, 32b)	0.5284	1.2115	0.5195%	13.29 сек	
C	LSTM (32u, 18w, 100e, 64b)	0.5950	1.2287	0.5857%	12.43 сек	
C	LSTM (32u, 18w, 200e, 16b)	0.6502	1.2068	0.6401%	26.15 сек	
C	LSTM (32u, 18w, 200e, 32b)	0.6395	1.2252	0.6298%	21.05 сек	
C	LSTM (32u, 18w, 200e, 64b)	0.5216	1.2117	0.5128%	18.57 сек	
C	LSTM (32u, 18w, 50e, 16b)	0.5681	1.2243	0.5589%	11.41 сек	
C	LSTM (32u, 18w, 50e, 32b)	0.5514	1.2208	0.5424%	16.69 сек	
C	LSTM (32u, 18w, 50e, 64b)	0.5523	1.2193	0.5432%	9.99 сек	
C	LSTM (32u, 24w, 100e, 16b)	0.6101	1.1693	0.6004%	17.49 сек	
C	LSTM (32u, 24w, 100e, 32b)	0.5900	1.2274	0.5807%	14.54 сек	
C	LSTM (32u, 24w, 100e, 64b)	0.5684	1.2241	0.5593%	13.05 сек	
C	LSTM (32u, 24w, 200e, 16b)	0.6363	1.2591	0.6264%	27.34 сек	
C	LSTM (32u, 24w, 200e, 32b)	0.5791	1.1706	0.5698%	21.96 сек	
C	LSTM (32u, 24w, 200e, 64b)	0.6115	1.2200	0.6020%	19.73 сек	
C	LSTM (32u, 24w, 50e, 16b)	0.5850	1.2268	0.5757%	12.78 сек	
C	LSTM (32u, 24w, 50e, 32b)	0.5835	1.2245	0.5743%	11.10 сек	
C	LSTM (32u, 24w, 50e, 64b)	0.5452	1.2193	0.5362%	9.99 сек	
C	LSTM (32u, 3w, 100e, 16b)	0.5673	1.2226	0.5581%	13.39 сек	
C	LSTM (32u, 3w, 100e, 32b)	0.5757	1.2239	0.5666%	14.41 сек	
C	LSTM (32u, 3w, 100e, 64b)	0.5696	1.2223	0.5604%	11.72 сек	
C	LSTM (32u, 3w, 200e, 16b)	0.6144	1.2316	0.6048%	18.88 сек	
C	LSTM (32u, 3w, 200e, 32b)	0.6530	1.2413	0.6432%	17.71 сек	
C	LSTM (32u, 3w, 200e, 64b)	0.5636	1.2220	0.5545%	15.81 сек	
C	LSTM (32u, 3w, 50e, 16b)	0.5329	1.2184	0.5240%	11.21 сек	
C	LSTM (32u, 3w, 50e, 32b)	0.5485	1.2194	0.5395%	10.67 сек	
C	LSTM (32u, 3w, 50e, 64b)	0.5696	1.2227	0.5605%	10.84 сек	
C	LSTM (32u, 6w, 100e, 16b)	0.6219	1.2345	0.6123%	13.33 сек	
C	LSTM (32u, 6w, 100e, 32b)	0.5731	1.2253	0.5639%	11.80 сек	
C	LSTM (32u, 6w, 100e, 64b)	0.5814	1.2266	0.5722%	17.44 сек	
C	LSTM (32u, 6w, 200e, 16b)	0.5644	1.2236	0.5552%	19.41 сек	
C	LSTM (32u, 6w, 200e, 32b)	0.6342	1.2374	0.6246%	16.95 сек	
C	LSTM (32u, 6w, 200e, 64b)	0.5761	1.2256	0.5669%	15.80 сек	
C	LSTM (32u, 6w, 50e, 16b)	0.5837	1.2271	0.5745%	10.23 сек	
C	LSTM (32u, 6w, 50e, 32b)	0.6183	1.2342	0.6088%	9.87 сек	

C	LSTM (32u, 6w, 50e, 64b)	0.5744	1.2239	0.5652%	9.18 сек	
C	LSTM (64u, 12w, 100e, 16b)	0.5769	1.2133	0.5677%	26.02 сек	
C	LSTM (64u, 12w, 100e, 32b)	0.6169	1.2220	0.6074%	18.12 сек	
C	LSTM (64u, 12w, 100e, 64b)	0.5672	1.2076	0.5581%	14.14 сек	
C	LSTM (64u, 12w, 200e, 16b)	0.5962	1.2191	0.5869%	24.22 сек	
C	LSTM (64u, 12w, 200e, 32b)	0.5328	1.2095	0.5239%	23.11 сек	
C	LSTM (64u, 12w, 200e, 64b)	0.5419	1.1991	0.5330%	19.18 сек	
C	LSTM (64u, 12w, 50e, 16b)	0.5449	1.2039	0.5360%	11.32 сек	
C	LSTM (64u, 12w, 50e, 32b)	0.5415	1.2196	0.5326%	10.23 сек	
C	LSTM (64u, 12w, 50e, 64b)	0.5648	1.2206	0.5557%	10.21 сек	
C	LSTM (64u, 18w, 100e, 16b)	0.5546	1.1868	0.5455%	16.87 сек	
C	LSTM (64u, 18w, 100e, 32b)	0.5733	1.2094	0.5641%	13.86 сек	
C	LSTM (64u, 18w, 100e, 64b)	0.5484	1.2127	0.5394%	13.36 сек	
C	LSTM (64u, 18w, 200e, 16b)	0.6735	1.2120	0.6628%	26.92 сек	
C	LSTM (64u, 18w, 200e, 32b)	0.6025	1.1842	0.5931%	20.85 сек	
C	LSTM (64u, 18w, 200e, 64b)	0.6470	1.2165	0.6372%	20.07 сек	
C	LSTM (64u, 18w, 50e, 16b)	0.5671	1.2104	0.5580%	14.95 сек	
C	LSTM (64u, 18w, 50e, 32b)	0.5939	1.2280	0.5846%	14.05 сек	
C	LSTM (64u, 18w, 50e, 64b)	0.5738	1.2226	0.5647%	10.37 сек	
C	LSTM (64u, 24w, 100e, 16b)	0.6379	1.1887	0.6279%	19.60 сек	
C	LSTM (64u, 24w, 100e, 32b)	0.5709	1.2114	0.5617%	14.99 сек	
C	LSTM (64u, 24w, 100e, 64b)	0.5908	1.2195	0.5815%	13.53 сек	
C	LSTM (64u, 24w, 200e, 16b)	0.6416	1.2519	0.6313%	29.39 сек	
C	LSTM (64u, 24w, 200e, 32b)	0.5959	1.1816	0.5866%	22.54 сек	
C	LSTM (64u, 24w, 200e, 64b)	0.6516	1.1831	0.6415%	20.03 сек	
C	LSTM (64u, 24w, 50e, 16b)	0.5440	1.2120	0.5351%	12.81 сек	
C	LSTM (64u, 24w, 50e, 32b)	0.5438	1.2203	0.5349%	11.83 сек	
C	LSTM (64u, 24w, 50e, 64b)	0.5507	1.2198	0.5417%	11.34 сек	
C	LSTM (64u, 3w, 100e, 16b)	0.5953	1.2279	0.5859%	13.74 сек	
C	LSTM (64u, 3w, 100e, 32b)	0.5633	1.2217	0.5542%	12.55 сек	
C	LSTM (64u, 3w, 100e, 64b)	0.5779	1.2236	0.5687%	11.95 сек	
C	LSTM (64u, 3w, 200e, 16b)	0.5743	1.2237	0.5651%	19.12 сек	
C	LSTM (64u, 3w, 200e, 32b)	0.6307	1.2355	0.6211%	17.24 сек	
C	LSTM (64u, 3w, 200e, 64b)	0.6297	1.2352	0.6201%	16.25 сек	
C	LSTM (64u, 3w, 50e, 16b)	0.5425	1.2193	0.5336%	9.97 сек	

C	LSTM (64u, 3w, 50e, 32b)	0.5327	1.2185	0.5238%	16.96 сек	
C	LSTM (64u, 3w, 50e, 64b)	0.5629	1.2212	0.5538%	10.40 сек	
C	LSTM (64u, 6w, 100e, 16b)	0.5672	1.2239	0.5580%	13.53 сек	
C	LSTM (64u, 6w, 100e, 32b)	0.6033	1.2307	0.5938%	12.55 сек	
C	LSTM (64u, 6w, 100e, 64b)	0.5789	1.2262	0.5697%	11.83 сек	
C	LSTM (64u, 6w, 200e, 16b)	0.5772	1.2257	0.5679%	20.88 сек	
C	LSTM (64u, 6w, 200e, 32b)	0.6086	1.2319	0.5992%	17.75 сек	
C	LSTM (64u, 6w, 200e, 64b)	0.5599	1.2227	0.5508%	16.97 сек	
C	LSTM (64u, 6w, 50e, 16b)	0.5660	1.2239	0.5569%	10.80 сек	
C	LSTM (64u, 6w, 50e, 32b)	0.6016	1.2304	0.5922%	9.83 сек	
C	LSTM (64u, 6w, 50e, 64b)	0.5872	1.2273	0.5780%	9.44 сек	




Среднее время обучения моделей
Ряд
C
3×10^1
2×10^1
10^1
Время (сек)
model

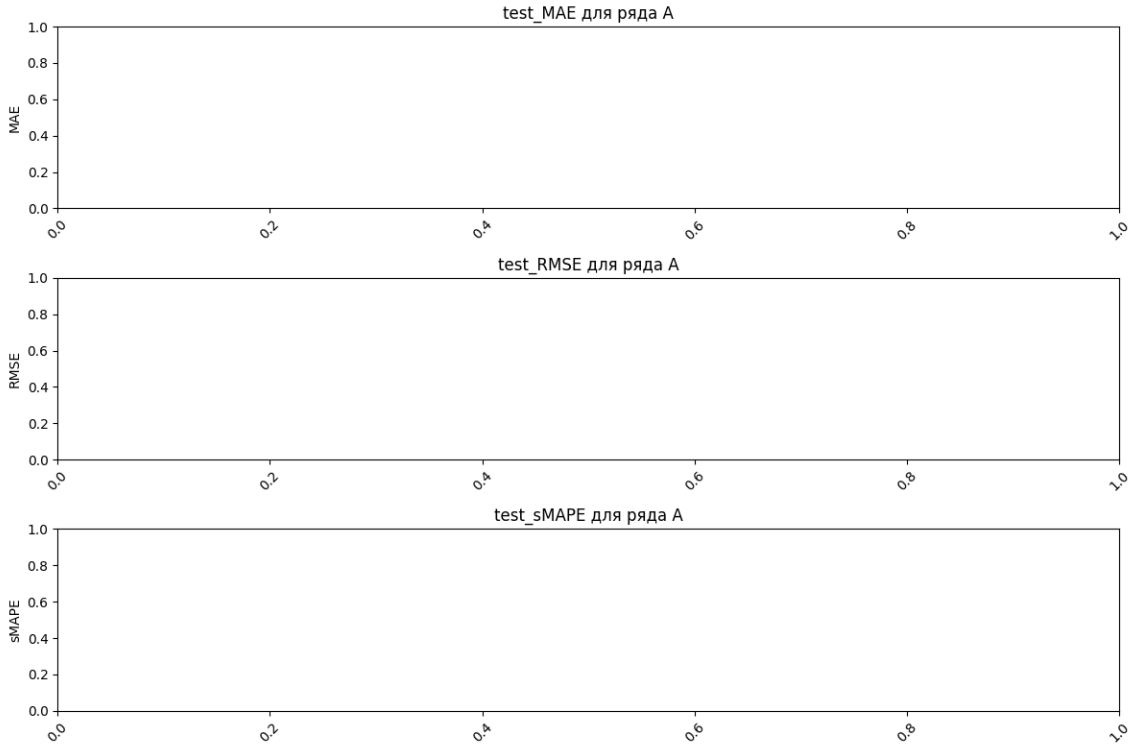
Анализ ряда A

Лучшая модель:

0
A
GRU (16u, 24w, 150e), 16b)
11.704305171966553
0.3123993601078594
0.33180706786782527
12.217951856981255
0.5437099530568105
0.5643125003292365
20.320353423394945
False

Детали модели:

Лучшая модель для ряда A: GRU (16u, 24w, 150e), 16b)
Тестовые метрики:
MAE: 0.5437
RMSE: 0.5643
sMAPE: 20.3204%
Время обучения: 11.70 сек


test_MAE для ряда A
MAE
test_RMSE для ряда A
RMSE
test_sMAPE для ряда A
sMAPE
0.0
0.2
0.4
0.6
0.8
1.0

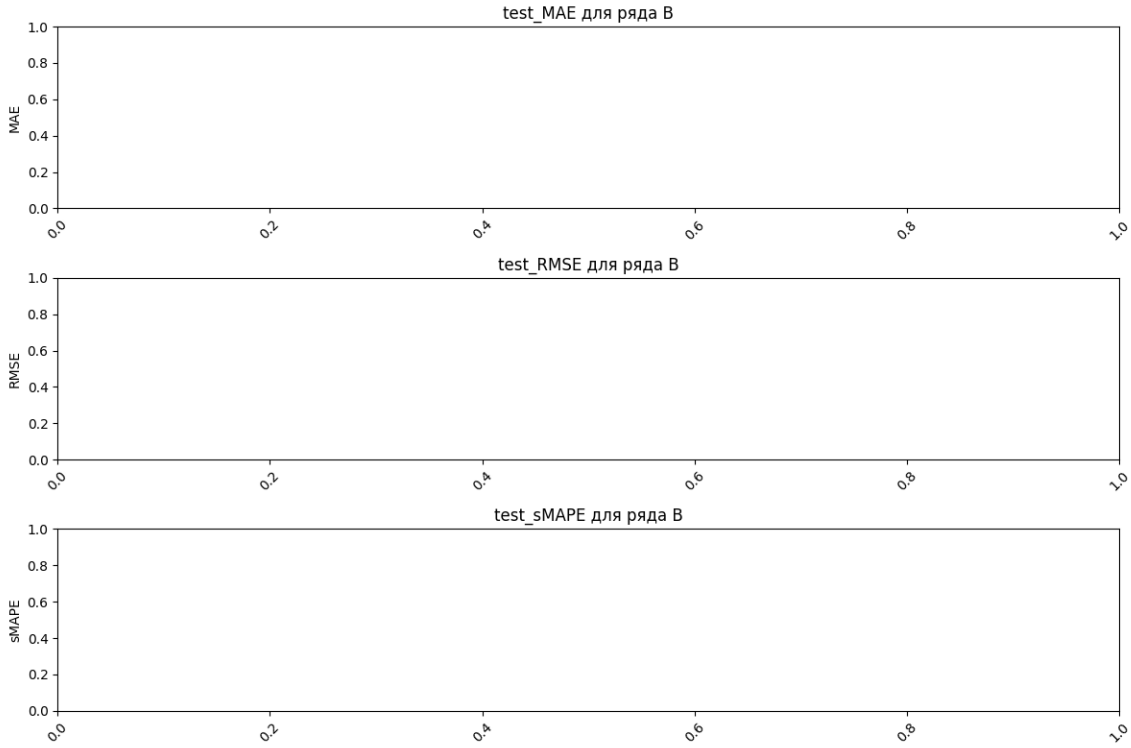
Анализ ряда B

Лучшая модель:

166
B
GRU (48u, 18w, 100e, 64b)
12.98051929473877
3.7713024616241477
4.374420919878364
3.7587790715472913
3.2698754975289996
4.020709377672101
3.2636380785920793
False

Детали модели:

Лучшая модель для ряда B: GRU (48u, 18w, 100e, 64b)
Тестовые метрики:
MAE: 3.2699
RMSE: 4.0207
sMAPE: 3.2636%
Время обучения: 12.98 сек


test_MAE для ряда B
MAE
test_RMSE для ряда B
RMSE
test_sMAPE для ряда B
sMAPE
0.0
0.2
0.4
0.6
0.8
1.0

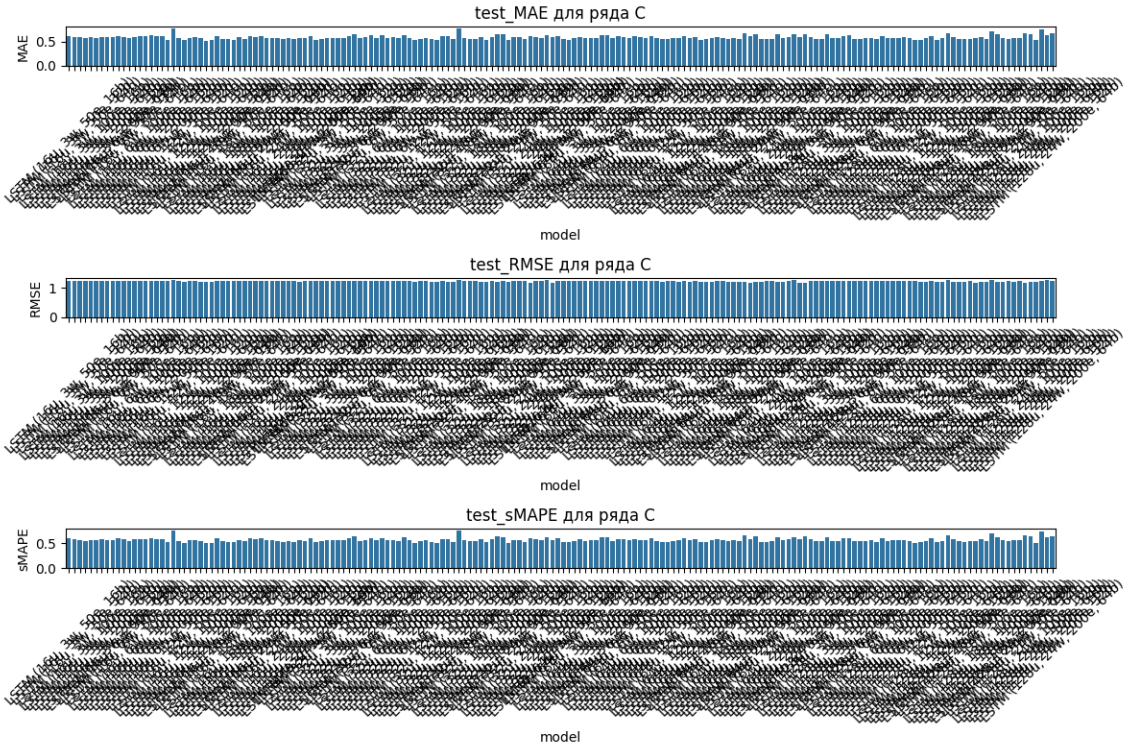
Анализ ряда C

Лучшая модель:

25
C
LSTM (16u, 12w, 200e, 32b)
23.633331537246704
0.34323439314558746
0.40143562195023125
0.3420001206840986
0.5104443359374983
1.2097711286919268
0.5016788252504116
False

Детали модели:

Лучшая модель для ряда C: LSTM (16u, 12w, 200e, 32b)
Тестовые метрики:
MAE: 0.5104
RMSE: 1.2098
sMAPE: 0.5017%
Время обучения: 23.63 сек


test_MAE для ряда C
MAE
0.5
0.0
model
test_RMSE для ряда C
RMSE
1
0
model
test_sMAPE для ряда C
sMAPE
0.5
0.0
model